

SOHAM KARMARKAR

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KEY COMPETENCIES

Languages: C++, Python, Java, HTML/CSS, JavaScript
Frameworks: BootStrap, Node JS, Flask, FastAPI
Database: MySQL, SQLite, MongoDB
ML: OpenCV, scikit-learn, NumPy, Pandas
Competitive Programming: 250+ DSA questions solved on LeetCode
Developer Tools: Git, VSCode, Tableau

EDUCATION

Sardar Patel Institute of Technology <i>Bachelor of Technology - Electronics and Telecommunication [CGPA 8.87/10]</i>	Mumbai, India Nov 2022 - Jun 2026
SP Jain Institute of Management and Research <i>Minor in Management</i>	Mumbai, India Jan 2024 - Nov 2025

PROJECTS

Spatia — GIS Tool for Spatial Operations and Visualisation <i>Java, GeoTools, JTS</i>	May 2025
- Developed a Java-based GIS application during HERE Hackathon 2025. - Designed a modular framework that supports 6+ core spatial operations and can scale to 10+ advanced GIS functions. - Enabled upload of .geojson, .shp, and .wkt files, performed operations (Union, Intersection, Difference, Buffer, Spatial Join), and rendered results on a 2D interactive map with zoom and pan.	
Laboratory Management & Optimisation System <i>React, FastAPI, MySQL, Roboflow</i>	Jan 2025 - Present
- Built a full-stack inventory system for hardware asset tracking across departments, reducing manual logging time by 75%. - Managed 500+ assets across labs with secure role-based access and RESTful APIs for real-time backend operations. - Integrated a custom-trained ML model (Electrocom 61 via Roboflow) for image-based classification of 50+ electrical components.	
Vehicle Detection in Aerial Imagery <i>YOLO models, LabelImg, Python</i>	Jan 2025
- Created and co-authored a research paper at ICMOCE 2025 (IIT Bhubaneswar), evaluating vehicle detection performance of YOLOv8n, YOLOv11n, and YOLOv12n on the SHIV-NATRAJ dataset. - Built a complete pipeline for frame extraction, annotation, YOLO-format conversion, training, and evaluation using metrics like Precision, Recall, F1-Score, mAP@50, and Fitness Score. - Benchmarked YOLOv8n for aerial surveillance, outperforming YOLOv11n by +3.4% precision, +4.6% recall, and +4.9% mAP@50, establishing it as the most accurate model for aerial surveillance performance	
Bank Loan Report Dashboard <i>Tableau, Excel, Data Analysis</i>	Oct 2024
- Analyzed 10,000+ loan applications, performing data cleaning to standardize metrics like loan status and funded amount. - Built 6 interactive visualizations (line charts, filled maps, donut charts, etc.) to track total loan applications, funded amount, and amount received. - Enabled cross-filtering by state, loan term, and employment length, identifying key trends in lending performance.	

ACHIEVEMENTS

Runner-up, HERE Technologies Hackathon	May 2025
Secured 2nd place at HERE Hackathon 2025 for developing Spatia, a GIS tool for uploading, analyzing, and visualizing geospatial data.	
Paper presented at ICMOCE Conference, IIT Bhubaneswar	May 2025
Co-authored and presented a research paper titled “Robust Vehicle Detection in Aerial Imagery: Comparative Study of YOLO Variants” at the 3rd ICMOCE Conference, to be included in IEEE proceedings.	

ROLES AND RESPONSIBILITIES

Chairperson, IEEE Computer Society SPIT	Aug 2024 - Present
Galvanized student interest in technical domains by spearheading monthly tech talks and coding competitions, attracting 750+ participants throughout tenure, while leading a team of 50 members.	
Head of Public Relations, IEEE Computer Society SPIT	Aug 2023 - May 2024
Maximized participation in events hosted by the committee and organized the Airavat AI Hackathon, attracting over 200 teams.	
Head of Operations, IEEE SPIT	Aug 2023 - May 2024
- Managed operations, coordinating logistics and resources to ensure smooth event execution. - Coordinated large-scale events, managing a logistics team of 15 and ensuring on-time execution.	